YEAR 9 PHYSICS · UNIT REVIEW

Work Like a Physicist

Unit review after Lesson 8: mixed exam-style questions from the whole unit, then a fix-it log for every lost mark.

"It doesn't matter how beautiful your theory is … If it doesn't agree with experiment, it's wrong."

Richard Feynman, physicist

NAME _____ CLASS _____

TEACHER _____ SCORE / 40 _____

ABOUT 45 MINUTES · USE THE TOOLKIT ON THE NEXT PAGE ONLY AFTER YOU HAVE TRIED EVERY QUESTION

REFERENCE

The Physicist's Toolkit

Everything you need for all eight lessons. Fold the corner of this page — you will come back to it.

The three variables

Independent	The one thing you change
Dependent	The thing you measure
Control	Everything you keep the same

Choosing your graph

Look at the independent variable only:

Continuous	Numbers on a scale — time, length, mass → **line graph**
Categoric	Names or groups — material, colour → **bar chart**

The two calculations

$$\text{mean} = \text{total of trials} \div \text{number of trials}$$

$$\text{uncertainty} = (\text{largest} - \text{smallest}) \div 2$$

Words that earn marks

Instead of "it went up", write "*as the ramp height increased, the distance travelled increased*". Instead of "it was wrong", name the error: *random* or *systematic*. Instead of "weird result", write "*possible anomalous result at 20 cm*".

Precision vs accuracy

Precise	Repeats are close *to each other*
Accurate	Close *to the true value*

You can be perfectly precise and completely wrong. That is the whole problem.

The two errors

Random	Scatters repeats unpredictably. **Repeats help.**
Systematic	Shifts every reading the same way. **Repeats do not help.**

Graph quality checklist

Title says what against what	☐
Independent variable on the x-axis	☐
Both axes labelled with units	☐
Sensible scale that fills the space	☐
Small crosses, plotted accurately	☐
Best-fit line, not dot-to-dot	☐
Bars equal width, with gaps	☐
Any odd point circled and explained	☐

THE RULE THAT DECIDES YOUR GRAPH

Continuous independent variable → line graph
Categoric independent variable → bar chart

It depends on what you *changed*, not on what you measured.

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UNIT REVIEW

Mixed exam practice

40 marks · about 45 minutes · show your working in every calculation

1 Timing a toy car [6 MARKS]

A student times how long a toy car takes to roll down a ramp from two different heights.

Ramp height / cm	Trial 1 / s	Trial 2 / s	Trial 3 / s	Mean / s
10	1.52	1.48	1.50	1.50
20	1.10	1.14	1.12	

a Calculate the mean time for the 20 cm ramp. Write it in the table. **[2]**

b Calculate the uncertainty in the mean time for the 20 cm ramp. Use: uncertainty = (largest − smallest) ÷ 2 **[2]**

c The stopwatch reads to 0.01 s. State its resolution. _____ **[1]**

d Suggest why using light gates would give more accurate times than a stopwatch. **[1]**

2 Keeping it warm [5 MARKS]

A student investigates how the **thickness of insulation** around a beaker of hot water affects the **temperature drop** in 10 minutes.

a Name the independent variable. _____ **[1]**

b Name the dependent variable. _____ **[1]**

c Name **two** control variables. **[2]**

d The thickness of insulation is [**categoric** | **continuous**] so the results should be shown on a [**bar chart** | **line graph**] **[1]**

3 Random or systematic? [4 MARKS]

a Tick **one** box in each row. [3]

Problem	Random error	Systematic error
A newtonmeter reads 0.5 N with nothing hanging on it.	☐	☐
The person timing reacts a little differently in each trial.	☐	☐
A scale is always read from slightly above, at the same angle.	☐	☐

b How should the student correct every reading from the newtonmeter? [1]

4 Precise or accurate? [4 MARKS]

Three students measure the length of a rod. Its true length is **50.0 cm**.

Student	Reading 1 / cm	Reading 2 / cm	Reading 3 / cm
A	49.9	50.1	50.0
B	48.2	48.3	48.2
C	47.0	52.5	50.4

a Whose readings are precise but **not** accurate? A B C [1]

b Whose readings are both precise and accurate? A B C [1]

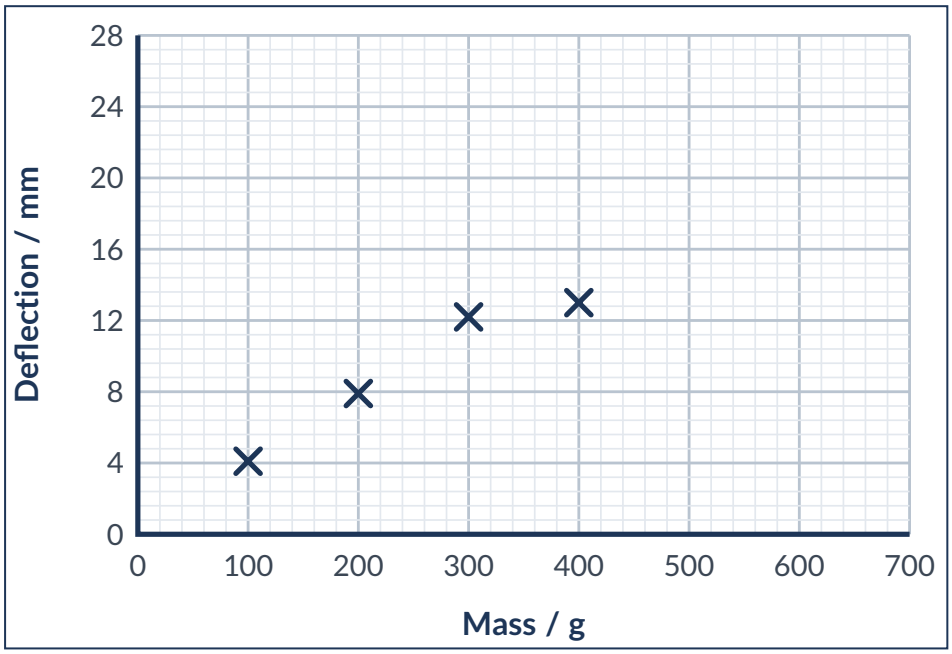
c The mean of C's readings is 49.97 cm, very close to the true length. Explain why C's readings are still not precise. [2]

5 A sagging beam

[9 MARKS]

A student hangs masses on the middle of a wooden beam and measures how far it sags (the deflection).

Mass / g	100	200	300	400	500	600
Deflection / mm	4.1	7.9	12.2	13.0	20.1	23.9



a Plot the results for 500 g and 600 g on the graph. [2]

b Draw a ring around the anomalous result. [1]

c Draw a line of best fit. [1]

d Use your graph to find the deflection for a mass of 250 g. _____ mm [1]

e What should the student do about the anomalous result? [1]

f Write a conclusion. Use the shape of the graph. [2]

g Explain why a line graph is the right choice for these results. [1]

6 Choosing a graph

[3 MARKS]

Tick **one** box in each row to show how the results should be presented.

Investigation	Bar chart	Line graph
How the type of ball affects its bounce height	☐	☐
How the temperature of a wire affects its resistance	☐	☐
How the colour of a surface affects its temperature rise in sunlight	☐	☐

7 Plan an investigation

[6 MARKS]

A student wants to find out how the **height** a ball is dropped from affects the **height it bounces**. Describe a method the student could use. Include how to make it a fair test and how to make the results repeatable.

8 Whose results can you trust? [3 MARKS]

Two groups did the bounce investigation. Group A took one reading for each drop height. Group B took three readings and calculated a mean.

a Explain why Group B's results are more trustworthy. [2]

b Group B publishes their results so that other scientists can check them. What is this called? [1]

✓ Fix-it log

TURN EVERY LOST MARK INTO A TARGET

Mark your answers with your teacher. For each mark you lost, write what went wrong and the correct physics, then the lesson to look back at.

Q	What went wrong?	The correct physics (or method) is…	Lesson to revisit

SCORE

My score: _____ / 40 Two things I will practise before the next test:

_____